JOURNAL OF MAGNETIC RESONANCE 96, 582–588 (1992)

NOTES

Parametric Estimation of Time-Domain NMR Signals Using Simulated Annealing

FRANK S. DIGENNARO AND DAVID COWBURN

The Rockefeller University, 1230 York Avenue, New York, New York 10021-6399

Received December 5, 1990; revised July 31, 1991

A limiting factor in accurately determining the parameters in nuclear magnetic resonance signals by various means has always been the signal-to-noise ratio (SNR) of the spectra. By definition of the Fourier transform pair

$$H(f) = \int h(t)\exp(2\pi ift)dt \qquad [1]$$

$$h(t) = \int H(f)\exp(-2\pi ift)df \qquad [2]$$

any noise sampled in the time domain is necessarily transformed into the frequency domain. This is not true for parametric estimation algorithms, where relevant, if imprecise, information can actually be extracted from the sampled (noisy) data. Linear prediction is a method used to extract information from time-domain data, but only if there is reasonably high SNR (*1*). In contrast, simulated annealing is not limited in this respect. Recently, this method has been applied to NMR signals to overcome the problem of low SNR in both the time domain (*2*) and the frequency domain (*3*). In this paper, we demonstrate that accurate results can be produced by this method and show in detail the working of the algorithm and its parameterization for practical NMR use.

Two basic algorithms used to minimize functions are (i) approximation algorithms and (ii) optimization algorithms. Approximation algorithms (also known as iterative improvement or neighborhood search algorithms) are usually very fast and yield a local or global minimum in polynomial time. Examples of iterative improvement algorithms include the method of steepest descents, the simplex method, and the various derivative searches (*4*). A disadvantage of this class of algorithms is that they search for and find the first minimum they stumble onto, regardless of whether it is local or global, and without any information as to how the final solution deviates from the optimal solution. The final solution then depends heavily on the initial configuration. To overcome this, one must usually repeat the optimization process for many initial configurations. It must be noted, however, that only in the limit as the number of initial configurations approaches infinity can one have strict assurance that the extremum found is truly the global one. Since this is not practical, optimization al-

0022-2364/92 $3.00
Copyright © 1992 by Academic Press, Inc.
All rights of reproduction in any form reserved.

gorithms such as the method of simulated annealing are used. It can be shown (*5*) that this method asymptotically finds a global minimum with probability 1 by using results from the theory of Markov chains, which are totally independent of initial conditions.

Annealing is a process in which a solid is heated sufficiently high to allow a random orientation of molecules and then is slowly and monotonically cooled. All molecules then arrange themselves in the lowest possible energy state, resulting in a very stable structure. As the temperature slowly approaches zero, only those states leading to the absolute minimum energy value have a nonzero probability of occurrence. The terms "sufficiently high" and "slowly cooled" are not well-defined terms and are system dependent. If we assume discrete temperature values, then, if for each temperature T the molecular system is allowed to reach thermal equilibrium, the probability that the system has an associated energy E is given by

$$P(E)dE \propto [\exp(-E/T)dT]/Z(T) \qquad [3]$$

$$1/Z(T) = \sum \exp(-E/T), \qquad [4]$$

where $\exp(-E/T)$ is the well-known Boltzman distribution, and $Z(T)$ is the normalization constant called the partition function. The natural log of $Z(T)$ is called the free energy. If the point of view is taken that annealing is simply an optimization process, where the free energy is the function to be minimized, this process could then be applied to other problems where one needs to find extreme points.

The first application of our algorithm was on several sets of synthetic data. This was done to optimize such variables as the initial value of the control parameter and the optimal Markov chain length. Before proceeding any further, a Markov chain should be defined. Consider a system that can exist in any one of N possible states. Let $x_0, x_1, \ldots, x_i, \ldots, x_N$ denote the allowed states, where x_i is the state following the ith transition. If, for all possible states $x_0, \ldots, x_N$, the probability that the system is in the x_{n+1} state is given by the equation

$$P(x_{n+1} \mid x_0, \ldots, x_n) = P(x_{n+1} \mid x_n), \qquad n = 0, 1, 2, \ldots, N, \qquad [5]$$

then the system is called a Markov chain. In other words, the probability of a transition from the x_n to the x_{n+1} state depends only on the nth state of the system and not on its history (or, more importantly, on its initial state x_0). In our case, the Markov chain length used was 600 times the number of spectral components.

We also needed to define the termination point of the routine, that is, how to define equilibrium in a precise way, and also to devise a generating mechanism to decrement the control parameter. Realizing that the execution time of such optimization routines generally increases nonlinearly with the size of the problem, in this case the number of peaks in the spectrum, it was a necessity to find the optimal values of these variables. Depending on the values of these variables, the convergence time varied by more than 300-fold.

In searching for the best initial value of the control parameter, we found that optimal results were obtained when we increased its value to the point where we found an 80–85% acceptance rate. This is analogous to "heating up the system" in an annealing sense. The temperature values were generally found to be inversely dependent on the

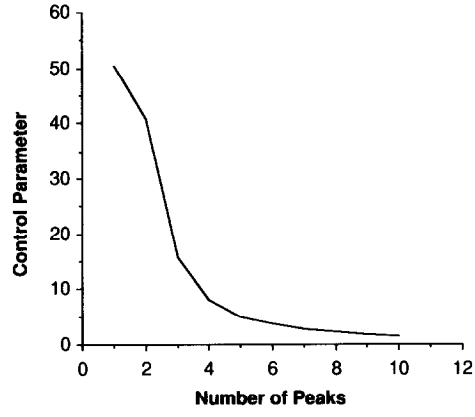


FIG. 1. Graph showing the initial value of the control parameter exceeding an 80% acceptance rate as a function of the number of peaks.

number of spectral components, as can be seen in Fig. 1. With regard to decrementing the control parameter, which can be thought of as a "cooling schedule," one would normally want the decrements to be small. In order to avoid long Markov chains in order to re-establish equilibrium for each new temperature value, a small decrement is normally used, e.g., $C_n = 0.9C_{n-1}$ (5). However, it was of interest to find out that if we begin to restrict the domain of all possible configurations, say after 10–20 temperature cycles, the control parameter can be decremented by a larger amount, e.g., 70%, each time and the same or better results be obtained with approximately 500% speedup in computation time. Without the restriction of ranges, such a cooling schedule is not usable, the difference in the error function being on the order of 10^6 or greater. With regard to the cooling schedule, we saw that if we strictly used an iterative algorithm (by setting $C = 0$), we fell into local minima relatively quickly, usually in less than 10–15 temperature cycles. Intrinsic to the annealing algorithm, we have a large number of acceptances by the Metropolis criterion initially, with the number approaching 0 as C approaches 0. Therefore, while the criterion is imperative earlier on, it can actually become counterproductive as C approaches 0, with our data showing that the error function can diverge by as much as a factor of 10 in the later stages without recovering. To overcome this, two things can be done. The first would be to use a more elaborate cooling schedule, i.e., decrement C more as C approaches 0. The preferred alternative is, after 10–15 cycles, set $C = 0$, effectively changing the algorithm over to an iterative improvement one, forcing a very rapid convergence. The point at which the "temperature" is set to 0 has appropriately been named the quenching point of the system. It is this cooling schedule that makes this current algorithm novel.

Another time-saving step was in our definition of equilibrium. Conventionally, the algorithm is terminated when no further improvement can be made to the estimate. In our case, the routine is terminated when the error function decreases below a certain threshold. It was found that no real benefit was gained if the algorithm continued beyond $R_e < 10^{-5}$. Continuing to $R_e < 10^{-10}$ required a 10-fold increase in computational time. To make the above section clearer, the algorithm is given below.

Program: Simulated Annealing

```
BEGIN
    read data
**calculate initial value of control parameter**
  REPEAT
    perturb(config i → config j)
    compute ΔR_e
    IF ΔR_e ≤ 0 OR exp(−ΔR_e/C) > random(0, 1) THEN accept
    IF accept THEN update variables
    compute acceptance ratio
    IF acceptance ratio <80% THEN C = 1.5C
  UNTIL success ratio ≥80%
**main**
  M = 0
  REPEAT
    REPEAT
      perturb(config i → config j),
      compute ΔR_e
      IF ΔR_e ≤ 0 OR exp(−ΔR_e/C) > random(0, 1) THEN accept
      IF accept THEN update variables
    UNTIL (Markov chain is finished or the maximum number of attempts have
        been reached)
  IF (quench point = true) THEN
    C_m = 0
    decrease ranges of parameters
  ELSE
    C_m = 0.3C_{m-1}
  UNTIL equilibrium point has been reached
END.
```

The program source was written in FORTRAN 77 and programs were compiled and run on a Stellar GS1000 benchmarked at 10 Linpack megaflops. Times for execution shown are elapsed times. Code is available from D.C. upon request.

Our algorithm was tested on several groups of data sets, the first group being several synthetic data sets, each containing a different number of spectral components with no noise (Table 1). This first group of data sets was used to test the functionality of the algorithm, that is, to show that the algorithm did indeed work properly under ideal conditions. The second group of data (Table 2) consisted of a data set having a more realistic SNR of about 5:1.

Lastly, a time-dependant vector of a 3D ^{15}N HMQC-NOESY spectrum of the protein adenylate kinase was processed. The vector was 128 points long, and the elapsed time to process was 189 s. The frequency-domain spectra of the Fourier and simulated-annealing results are shown in Fig. 2.

Simulated annealing would appear to be a potentially useful tool for the component analysis of NMR signals. It was seen that the algorithm found the optimal parameters

TABLE 1

	Input	Output
	Data set 1: One peak	
Amplitude	1.00	1.00
Frequency (Hz)	40.00 Hz	39.98
Line broadening (Hz)	0.2 Hz	0.20
Phase (degrees)	0.00	0.00
Spectral width (Hz)	1000	
FID length (points)	16	
Time (s)	3	
R_e	2×10^{-5}	
	Data set 2: Five peaks	
Amplitude	5, 3, 3, 4, 5	5.00, 3.00, 3.00, 4.00, 5.01
Frequency (Hz)	40, 200, 300, 400, 500	40.01, 200.01, 300.01, 400.01, 499.99
Line broadening (Hz)	0.2, 0.2, 0.3, 0.4, 0.5	0.20, 0.20, 0.30, 0.39, 0.49
Phase (degrees)	0, 0, 0, 0, 0	1.28, 1.28, 0.28, 1.18, 0.00
FID length (points)	64	
Time (s)	85	
R_e	9×10^{-7}	
	Data set 3: Seven peaks	
Amplitude	5, 3, 3, 4, 5, 4, 3	4.99, 2.99, 2.99, 4.00, 5.01, 3.01
Frequency (Hz)	40, 200, 300, 400, 500, 600, 750	39.98, 200.03, 300.03, 400.04, 500.02, 600.02, 750.04
Line broadening (Hz)	0.2, 0.2, 0.3, 0.4, 0.5, 0.6, 0.4	0.20, 0.20, 0.31, 0.40, 0.49, 0.59, 0.39
Phase (degrees)	0, 0, 0, 0, 0, 0, 0	0.00, 1.28, 0.28, 0.28, 1.08, 1.28, 2.28
FID length (points)	128	
Time (s)	175	
R_e	8×10^{-6}	

in a systematic order. The frequencies were found first, then the amplitudes and phases, and then finally the line broadenings, with a very large percentage of time being used finding the frequencies. If the frequencies were found, then all other parameters followed. When the frequencies were initially set constant, all other parameters were found with a speedup in computational time by a factor of two. If combined with linear predictive singular value decomposition (LPSVD) or Fourier transformation to find the frequencies, this technique might find more widespread use, especially for low SNR cases when LPSVD is of limited use for parametric estimation (*6*). The execution times for simulated annealing are currently prohibitive for most uses, and because of this limitation we have not, in this article, established the algorithm's practicality for many larger NMR problems to which it might be applied. Like LPSVD,

TABLE 2

	Input	Output
	Data set 4: Four peaks, SNR 5:1	
Amplitude	1, 2, 1, 2	1.04, 2.02, 1.03, 1.99
Frequency (Hz)	40, 200, 350, 450	40.17, 199.97, 350.01, 449.93
Line broadening (Hz)	0.20, 0.20, 0.30, 0.40	0.19, 0.20, 0.28, 0.48
Phase (degrees)	0, 0, 0, 0	1.25, 0.00, 1.26, 0.00
FID length (points)	128	
Time (s)	235	

this simulated-annealing approach reduces time-varying observations to components directly, and with the similar nominal advantages of LPSVD (*7*), provides resolution unobtainable by enhancement/Fourier methods, achieves the elimination of *sinc* modulation caused by signal truncation with zero filling/Fourier procedures, removes the need for phase-sensitive detection resulting from t_1 phase modulation in 2D experiments, and it can be used to facilitate automated extraction of chemical information from the resulting list of components. In addition, simulated annealing readily permits analysis with increased prior information like frequency ranges, limited decay constants, or restricted phases [cf. Ref. (*8*)]. Lastly, simulated annealing is an attractive algorithmic route for problems in connectivity and assignment (like the "traveling salesman" problem), and the combination of these procedures with component estimation may be useful. The long time for calculation may be substantially relieved by advances in large-scale parallel processing; LPSVD was similarly restricted by speed of calculation, but has shown considerable recent applicability (*9*).

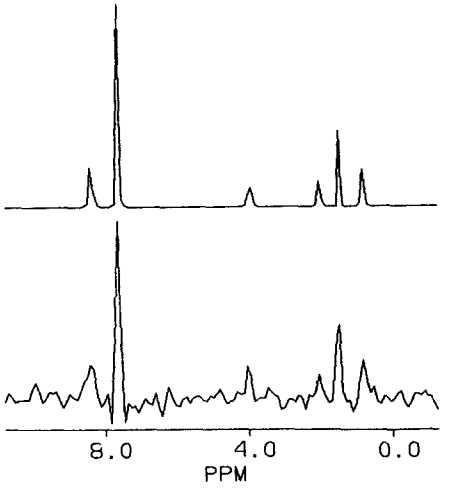


FIG. 2. Fourier-transformed data set from a 3D HMQC-NOESY experiment (bottom) and the results after it has been annealed (top) [see Ref. (6, Fig. 4) for exact conditions].

ACKNOWLEDGMENTS

We thank C. B. Rios for providing the 3D adenylate kinase data. This work was supported in part by NIH Grant DK-20357.

REFERENCES

1. R. KUMARESAN AND D. W. TUFTS, *IEEE Trans. Acoust. Speech Signal Process.* **ASSP-30,** 833 (1982).
2. K. SEKIHARA AND N. OHYAMA, *Magn. Reson. Med.* **13,** 332 (1990).
3. R. E. HOFFMAN AND G. C. LEVY, *J. Magn. Reson.* **83,** 411 (1989).
4. W. H. PRESS, B. P. FLANNERY, S. A. TEUKOLSKY, AND W. T. VETTERLING, "Numerical Recipes," Cambridge Univ. Press, New York, 1989.
5. P. J. M. VAN LAARHOVEN AND E. H. L. AARTS, "Simulated Annealing: Theory and Applications," Reidel, Dordrecht, 1987.
6. D. COWBURN, J. GLUSHKA, F. S. DIGENNARO, AND C. B. RIOS, "Proceedings, NATO Conference of Computational Aspects of the Study of Biological Macromolecules by NMR," in press.
7. A. SCHUSSHEIM AND D. COWBURN, *J. Magn. Reson.* **71,** 371 (1987).
8. G. ZHU AND A. BAX, *J. Magn. Reson.* **90,** 405 (1990).
9. G. M. CLORE, L. E. KAY, A. BAX, AND A. M. GRONENBORN, *Biochemistry* **30,** 12 (1991).